

Gaurav Bhatia

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Portfolio | GitHub | LinkedIn

Junior Machine Learning Engineer

MSc Data Science candidate at GISMA, finishing December 2026, who ships models as services. Builds with **Python, scikit-learn, XGBoost and TensorFlow**, tunes against hold-out data (**50 Optuna trials**, MAE 22.98 Wh), and serves them through **FastAPI and Docker**. Evaluates model variants at KnowGuard AI, and is building ParetoGate, an LLM gateway with ten trained classifiers. Available full-time in Berlin from **December 2026**.

Skills

Machine learning	scikit-learn, XGBoost, TensorFlow/Keras (LSTM), Optuna hyperparameter tuning, SHAP explainability, time-series (ARIMA, Prophet, Chronos), model evaluation (MAE, RMSE, MAPE), A/B testing
Programming	Python (pandas, NumPy, statsmodels, scikit-learn, XGBoost, Optuna, TensorFlow/Keras), SQL (CTEs, window functions), DuckDB, Snowflake SQL, SQLite
Serving & delivery	FastAPI, REST APIs, Docker, Kubernetes, GitHub Actions CI/CD, Git/GitHub, AWS (A/B testing workloads), reproducible and documented pipelines
AI engineering	LLM evaluation and quality gates, guardrails, RAG (LangChain, FAISS), agentic workflows, semantic caching and model routing, LLM cost optimisation, human-in-the-loop design, MiniLM embeddings, Hugging Face
Languages	English (C1), German (A2, improving), Hindi (native)

Experience

Data Science Intern — Aug 2026 – Present
KnowGuard AI — *Berlin, Germany*

- Build **RAG pipelines** and **multi-agent systems**, and define evaluation metrics and hold-out sets so model comparisons are reproducible rather than anecdotal
- Ran a **paired A/B test on AWS** comparing a fine-tuned model against the review arm: **+0.0318 AUPRC** (95% CI 0.022–0.0418), below the pre-set 0.05 practical-effect threshold
- **Benchmark LLMs against baselines** and design **cost-effective evaluation workflows** for model selection

Operations & Supply Chain Analyst — Mar 2024 – Jul 2025
Troika International Pvt. Ltd. — *Panchkula, India*

- Monitored inventory and demand across **700+ vendors** and **500+ SKUs** in Tally and Excel
- Built **Power BI dashboards across 3 divisions**, turning operational data into clear views for non-technical stakeholders
- Rescued a stalled **₹2.75M export order** (500 units to Lithuania) by moving the shipment to **air freight** to hold the delivery window and renegotiating terms with the client

Business Manager — Nov 2015 – Jan 2024
Bhatia Traders — *India*

- Ran a product business end to end for **8 years**, owning purchasing, pricing, stock and supplier terms across **20+ suppliers**
- Analysis-led changes cut **excess-inventory cost by 10%** and identified a ~**15% stockout-reduction** potential

Education

GISMA University of Applied Sciences — Potsdam, Germany
MSc Data Science, AI & Digital Business — *Sep 2025 – Dec 2026 (expected)*

- Thesis: the US–Iran conflict's impact on German day-ahead electricity prices, using event study, Granger causality and Chronos/XGBoost forecasting
- Relevant modules: Methods of Prediction (M507), Big Data Analytics (M508)

University College of Engineering, Patiala — *B.Tech, Civil Engineering, 2011 – 2015*

Selected Projects

ParetoGate: cost-aware LLM gateway — *FastAPI, MiniLM, scikit-learn, SQLite, Ollama* — Private repo

- Built a local-first **LLM gateway** that classifies each request with MiniLM embeddings, predicts applicable optimisations with ten one-vs-rest logistic models, and applies deterministic safeguards before any lossy transform
- Cut retrieved context from **86,416 to 11,351 input tokens** on a frozen 25-task code-change suite while passing every reviewed change, an **80.8% modelled cost reduction**
- Gated every saving behind a **task-quality floor** with automatic baseline fallback, charging rejected candidates as zero savings; reported a negative end-to-end run rather than publishing the favourable 81% token figure alone

Sales Intelligence Agent — *TF-IDF + logistic regression, Gemini, FastAPI, Streamlit* — Source · Demo

- Routed plain-English business questions through a local **intent classifier**, calling Gemini only below a confidence threshold to control cost
- Served it as a **FastAPI microservice** with a Streamlit front end and versioned training and hold-out sets, so accuracy stays measurable

RAG Retail Agent *LangChain, FAISS, FastAPI, Docker, Kubernetes* Source

- Built a **LangChain + FAISS** retrieval pipeline over product data behind a **FastAPI** service, with **GitHub Actions to Docker Hub to Kubernetes** CI/CD

Office Energy Forecasting Dashboard *XGBoost, Optuna, SHAP, Streamlit* Source · Demo

- Tuned XGBoost with **50 Optuna trials** to predict office energy use to **MAE 22.98 Wh** and RMSE 50.96 Wh, surfaced with CO_2 savings in a live Streamlit app
- **SHAP** showed short-term momentum dominates consumption, driving out-of-hours waste alerts

Solar Power Forecasting (LSTM) *TensorFlow/Keras, OPSD, Open-Meteo* Source

- Forecast hourly German solar generation with an **LSTM** on **26,280 hourly rows**, using a chronological 70/15/15 split to prevent leakage
- Evaluated on held-out data with RMSE, MAE and MAPE

MSc Thesis: Conflict Shocks in the German Power Market *Python, Chronos, XGBoost, Granger causality* Source

- Built a reproducible pipeline answering **5 research questions** on how the 2026 US–Iran conflict moved German day-ahead electricity prices, benchmarked against the 2022 energy shock
- Combined **event study**, **Granger causality** and ablation tests with **Chronos** and XGBoost forecast arms to identify the transmission channel

Demand Forecasting Engine *Python, ARIMA, Prophet, XGBoost, Streamlit* Source

- Compared ARIMA, Prophet and XGBoost on 5 years of weekly sales for 12 SKUs on a 12-week hold-out; **Prophet best at 6.07% MAPE** vs. 15.49% for ARIMA
- Shipped as a Streamlit app where a planner picks SKU, model and horizon and sees MAE/RMSE/MAPE live

Dynamic Pricing & Segmentation *Python, statsmodels, XGBoost, Plotly* Source

- Estimated **price elasticity per category** with log-log OLS on 15,000 SKUs in 8 categories (e.g. Fashion −2.1, Food & Beverages −0.5)
- Segmented customers with **RFM scoring**; 6 business rules + XGBoost repriced **55.7% of the catalogue** (8,352 SKUs), and an XGBoost price recommender reached **2.14% MAPE**

Job-Search Automation: agent harness *Scheduled agents, shared database, Python, GitHub* Private repo

- Built a versioned system of **four scheduled agent tasks** (sourcing, outreach, daily report, LinkedIn drafts) that hand work to each other through a **shared database**
- Sourced roles from ATS and Arbeitsagentur feeds, government portals and job boards across **12 countries**, plus funding news for startups

Energy News and Price Volatility (NLP) *FinBERT, VADER, TF-IDF, XGBoost, SHAP* Source

- Predicted high-volatility days on EPEX Spot from **4,444 energy headlines** using FinBERT and VADER sentiment, TF-IDF trend signals and Google Trends attention
- Compared **four models** (dummy, FinBERT-only, default and tuned XGBoost) with SHAP attributions and Granger-causality-informed lags

Certifications

- **Introduction to Data Engineering with Microsoft Azure 1 & 2**, FutureLearn, Sep 2026
- **Introduction to Cloud Infrastructure: Describe Cloud Concepts**, Microsoft, Jul 2026
- **Data Analytics Job Simulation**, Deloitte, issued by Forage, Jul 2026